Complete Cotton (Gossypium spp.) Agronomy, IPM & Production Manual - Page 1

1. Taxonomy & Cultivated Species

- Gossypium hirsutum (Upland/American Cotton): Tetraploid (2n=52); accounts for >90% of global commercial production.
- Gossypium barbadense (Egyptian/Pima/Sea Island): Tetraploid (2n=52); prized for extra-long staple (ELS) luxury fibers.
- Gossypium arboreum (Tree Cotton) & G. herbaceum (Levant Cotton): Diploids (2n=26); drought & pest tolerant short-staple.

2. Edaphic (Soil) & Climatic Requirements

- Soil Type: Deep, friable, well-drained black cotton soils (Vertisols), alluvial, or red loamy soils with good moisture retention.
- Soil pH & Salinity: Optimal pH 6.0 to 8.0; moderately tolerant to salinity (EC threshold ~7.7 dS/m), highly sensitive to waterlogging.
- Climate: Requires 180-200 frost-free days, abundant sunshine. Germination: >16 C; Vegetative: 21-27 C; Fruiting: 27-32 C day / cool nights.

3. Crop Phenology & Growth Stages (Days After Sowing - DAS)

- Emergence to Seedling (0-30 DAS): Epigeal germination in 5-10 days; deep taproot establishment; high vulnerability to thrips/aphids.
- Squaring (35-55 DAS): Appearance of floral buds (squares); rapid nodal development and canopy expansion.
- Peak Flowering to Boll Formation (60-100 DAS): White flowers turn pink then shed petals; peak water and potassium demand.
- Boll Maturation & Dehiscence (110-160+ DAS): Fiber elongation (days 1-20 post-anthesis) and secondary wall thickening (days 20-45).

4. Land Preparation, Seed Treatment & Sowing Agronomy

- Tillage: Deep summer plowing (20-30 cm) with moldboard plow every 2-3 years to expose pupae and perennial weeds (Cyperus rotundus).
- Acid Delinting: Treat fuzzy seed with concentrated H2SO4 (100 ml/kg seed) for 2-3 mins, wash thoroughly, and lime-neutralize.
- Seed Dressing (per kg): Imidacloprid 70 WS (5-7 g) for sucking pests + Carboxin 75 WP (2 g) + Trichoderma viride (10 g) for root rot.
- Seed Rate & Spacing: Bt Hybrids: 1.5-2.5 kg/ha at 90-120 x 45-60 cm. HDPS (High Density Planting System): 10-12 kg/ha at 60 x 10 cm.
- Refugia Mandate: Plant 5% non-Bt cotton (or 20% in bag-in-bag systems) to delay bollworm resistance to Cry1Ac and Cry2Ab toxins.

5. Integrated Nutrient Management (INM) & Physiological Disorders

- Organic Manuring: Incorporate 10-12 tonnes/ha of well-decomposed Farm Yard Manure (FYM) or vermicompost 2-3 weeks prior to sowing.
- NPK Dosage: Hybrids: 120-150:60:60 kg N:P2O5:K2O/ha. Varieties: 60-90:30:30 kg/ha. Apply 100% P & K + 25% N as basal dose.
- Top Dressing: Apply remaining 75% Nitrogen in 3 equal splits at 30 DAS (thinning), 60 DAS (squaring), and 90 DAS (peak flowering).
- Micronutrients: Soil apply ZnSO4 (25 kg/ha) and Borax (5-10 kg/ha) in deficient soils; Boron prevents hollow-heart and square shedding.
- Foliar Nutrition: Spray 2% KNO3 (13:0:45) or 2% DAP at 60, 75, and 90 DAS to boost boll weight and fiber micronaire.
- Leaf Reddening (Lalya): Caused by Mg deficiency and night temperature drops. Manage with 1% MgSO4 + 1% Urea foliar spray at 60-75 DAS.
- Tirak (Bad Boll Opening): Premature cracking with immature lint on saline/light soils. Mitigate via timely irrigation and K application.
- Physiological Square Shedding: Spray Naphthalene Acetic Acid (NAA / Planofix) at 10-20 ppm at flower initiation and 15 days later.

6. Irrigation Scheduling, Weed Control & Canopy Management

- Water Requirement: 700-1200 mm total. Critical moisture stages: Squaring (35-45 DAS), Flowering (60-75 DAS), Boll Development (85-105 DAS).
- Irrigation Methods: Furrow irrigation (alternate furrow saves 30% water) or Drip Fertigation (increases water-use efficiency by 40-50%).
- Pre-Emergence Weed Control: Spray Pendimethalin 30 EC (3.3 L/ha) or Pendimethalin 38.7 CS (1.75 L/ha) within 48 hours of sowing.
- Post-Emergence & Intercultivation: Hoeing at 25 and 45 DAS; directed spray of Pyrithiobac-sodium 10 EC (625 ml/ha) + Quizalofop-ethyl.
- Canopy Regulation: Detopping terminal bud at 90-100 DAS or spraying Mepiquat Chloride 5% AS (50 ppm / 1 ml/L) at 60 and 75 DAS.

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7. Comprehensive Integrated Pest Management (IPM) & Economic Threshold Levels (ETL)

A. Bollworm Complex (Lepidoptera)

- American Bollworm (Helicoverpa armigera): Larvae feed on squares/bolls with half body outside; circular bore holes with flared bracts. ETL: 1 larva/plant or 5-10% damaged fruiting bodies. Control: HaNPV (250-500 LE/ha), Emamectin benzoate 5 SG (0.4 g/L), Chlorantraniliprole.
- Pink Bollworm (Pectinophora gossypiella): Internal feeder; causes 'rosette flowers', interlocular burrowing, stained lint, ruined seeds. ETL: 10% rosette flowers or 8 moths/pheromone trap/night for 3 nights. Control: Gossyplure traps (20/ha), Profenofos 50 EC or Spinetoram.
- Spotted Bollworm (Earias vittella & E. insulana): Terminal shoot boring (drooping/withering) in early stages; lint-fouling frass in bolls. ETL: 5% withered shoots or damaged bolls. Control: Clip and destroy infested shoots; release Trichogramma chilonis (1.5 lakh/ha).
- Tobacco Caterpillar (Spodoptera litura): Gregarious larvae skeletonize leaves. Control: Castor trap crop; SlNPV; Novaluron 10 EC.

B. Sap-Sucking & Foliage Pest Complex

- Whitefly (Bemisia tabaci): Sucks phloem sap, excretes honeydew (sooty mold), transmits CLCuD. ETL: 6-8 adults or 20 nymphs/leaf. Control: Yellow sticky traps (25/ha); Neem oil 0.5%; Diafenthiuron 50 WP (1.2 g/L), Pyriproxyfen 10 EC (2 ml/L), or Spiromesifen.
- Leafhopper / Jassids (Amrasca biguttula biguttula): Causes downward leaf curling, yellowing, marginal reddening, and 'hopper burn'. ETL: 2-3 nymphs/leaf or Grade II injury. Control: Grow hairy varieties; spray Flonicamid 50 WG (0.3 g/L) or Dinotefuran 20 SG (0.3 g/L).
- Aphids (Aphis gossypii) & Thrips (Thrips tabaci): Aphids cause crinkling; Thrips lacerate leaves causing silvery patches and brittle curl. ETL: 10-15% infested plants (Aphids) or 10 thrips/leaf. Control: Conserve Coccinellids/Chrysoperla; spray Thiamethoxam or Spinosad.
- Mealybug (Phenacoccus solenopsis): Waxy colonies on stems cause stunted, bushy shoots. Control: Destroy Parthenium weeds; Buprofezin 25 SC.
- Red & Dusky Cotton Bugs (Dysdercus koenigii & Oxycarenus laetus): Stain lint yellow/red and reduce seed oil content and viability.

8. Plant Pathology & Disease Management

- Cotton Leaf Curl Disease (CLCuD): Begomovirus (Geminiviridae) vectored by Bemisia tabaci. Symptoms: Vein thickening, cup-shaped enations. Management: Eradicate weed hosts (Sida, Abutilon); sow tolerant hybrids; strictly avoid synthetic pyrethroids to prevent whitefly resurgence.
- Bacterial Blight / Blackarm (Xanthomonas citri pv. malvacearum): Water-soaked angular leaf spots, vein blight, black stem lesions, boll rot. Management: Acid delinting; spray Copper Oxychloride 50 WP (2.5 g/L) + Streptocycline (100 ppm) at 15-day intervals on symptom appearance.
- Fusarium Wilt (F. oxysporum f. sp. vasinfectum) & Verticillium Wilt (V. dahliae): Yellowing, wilting, dark brown vascular ring discoloration. Management: Potash application; seed & soil application of Trichoderma harzianum + Pseudomonas fluorescens; spot drench Carbendazim (1 g/L).
- Root Rot (Rhizoctonia bataticola / R. solani): Sudden wilting in patches; shredded bark on roots. Manage by intercropping with Mothbean/Sorghum.
- Grey Mildew (Ramularia areola) & Alternaria Leaf Spot: Frosty white growth on lower leaves or concentric brown rings. Spray Azoxystrobin + Tebuconaz

9. Defoliation, Harvesting, Post-Harvest & Fiber Quality Grading

- Defoliation & Boll Opening: For mechanical picking, apply Thidiazuron + Diuron (at 60% boll opening) and Ethephon 39 SL to synchronize boll opening.
- Harvesting Protocol: Pick dry seed-cotton (Kapas) in morning hours after dew evaporates; keep 3-4 pickings separate (first & last pickings are inferior).
- Post-Harvest Storage: Dry Kapas in shade to <8-10% moisture; store in clean cotton cloth bags (avoid polypropylene bags to prevent fiber contaminatior
- Ginning & Parameters: Ginning Outturn (GOT) is typically 32-36% lint and 64-68% seed. High Volume Instrument (HVI) grades: Upper Half Mean Length (UHML: Short <20 mm, Long 25-30 mm, Extra-Long >32 mm), Micronaire (fineness: optimal 3.5-4.9), Bundle Tenacity (>28 g/tex), and Trash Content.